

Existence and Uniqueness of the Stationary Distribution

<https://asanchowk.github.io/>

(This blogpost follows the exposition of [1], but in an idiosyncratic and verbose manner.)

Assume a *finite irreducible and aperiodic* Markov Chain $\{X_t\}_{t \geq 0}$ with transition matrix P . We prove, by construction, the existence of the stationary distribution π for every such Markov Chain.

We point the reader to [this post](#) for the notation used in this blog post. We specially point out the notation of first return time τ_z^+ for a random walk starting at $z \in \mathcal{X}$ and the element-wise formulation of the stationary distribution π for reference.

1 Setup and Overview

Let $z \in \mathcal{X}$ be any arbitrary state, and define

$$\begin{aligned}\tilde{\pi}(y) &= \mathbb{E}_z[\text{number of visits to } y \text{ before returning to } z] \\ &= \sum_{t=0}^{\infty} \Pr_z[X_t = y, \tau_z^+ > t]\end{aligned}$$

Our argument will broadly be in two steps:

1. We first show that $\tilde{\pi}P = \tilde{\pi}$.
2. $\tilde{\pi}$ is clearly not a probability distribution¹, so we say that $\pi = \frac{\tilde{\pi}}{\sum_{x \in \mathcal{X}} \tilde{\pi}(x)}$ is a stationary distribution on $\mathcal{X}$. We will later show that $\sum_{x \in \mathcal{X}} \tilde{\pi}(x) = \mathbb{E}_z[\tau_z^+]$, so the final stationary distribution we construct will be $\pi = \frac{\tilde{\pi}}{\mathbb{E}_z[\tau_z^+]}$.

It is essential to note that we omit one important part of the proof here in our exposition. We need to show that $\tilde{\pi}(y)$ cannot be infinite for any $y \in \mathcal{X}$. We omit this part of the proof and refer the reader to Lemma 1.13 of Levin-Peres-Wilmer book.

¹This could be seen as trivial in retrospect, but it is easy to forget that a vector π satisfying $\pi P = \pi$ shows the stationarity of π but doesn't imply that π is a probability distribution. For some (unrigorous) intuition, consider this. Recall that we have formulated π as a row vector - if it was a column vector instead, our $\pi P = \pi$ equation would look uncannily close to the $Px = \lambda x$ formulation we see often when discussing eigenvalues in linear algebra, and there is no restrictions that the elements of x sum to 1. I hope to explore these connections to spectral theory in a future post.

2 Proof of existence

Proposition 1. (Proof of existence) $\pi = \frac{\tilde{\pi}}{\mathbb{E}_z[\tau_z^+]}$ is a stationary distribution.

Proof. We begin by showing $\tilde{\pi}P = \tilde{\pi}$:

$$\begin{aligned}
\tilde{\pi}P &= \sum_{x \in \mathcal{X}} \tilde{\pi}(x) \cdot P(x, y) \\
&= \sum_{x \in \mathcal{X}} P(x, y) \sum_{t=0}^{\infty} \Pr_z[X_t = x, \tau_z^+ > t] && \text{(from the definition of } \tilde{\pi} \text{)} \\
&= \sum_{x \in \mathcal{X}} \sum_{t=0}^{\infty} \Pr_z[X_t = x, \tau_z^+ > t] \cdot P(x, y) && \left(M \cdot \sum x_i = \sum M \cdot x_i \text{ if } \sum x_i = 1 \right) \\
&= \sum_{t=0}^{\infty} \sum_{x \in \mathcal{X}} \Pr_z[X_t = x, \tau_z^+ > t] \cdot P(x, y) && \text{(switching order of summations)} \\
&= \sum_{t=0}^{\infty} \sum_{x \in \mathcal{X}} \Pr_z[X_t = x, \tau_z^+ \geq t+1] \cdot P(x, y) \\
&= \sum_{t=0}^{\infty} \sum_{x \in \mathcal{X}} \Pr_z[X_t = x, X_{t+1} = y, \tau_z^+ \geq t+1] \\
&= \sum_{t=0}^{\infty} \sum_{x \in \mathcal{X}} \Pr_z[X_t = x, X_{t+1} = y, \tau_z^+ \geq t+1] \\
&= \sum_{t=0}^{\infty} \sum_{x \in \mathcal{X}} \Pr_z[X_{t+1} = y, \tau_z^+ \geq t+1] \cdot \Pr_z[X_t = x] \\
&= \sum_{t=0}^{\infty} \Pr_z[X_{t+1} = y, \tau_z^+ \geq t+1] && \left(\sum_{x \in \mathcal{X}} \Pr_z[X_t = x] = 1 \right) \\
&= \sum_{t=1}^{\infty} \Pr_z[X_t = y, \tau_z^+ \geq t] \\
&= \sum_{t=1}^{\infty} \Pr_z[X_t = y, \tau_z^+ > t] + \sum_{t=1}^{\infty} \Pr_z[X_t = y, \tau_z^+ = t] \\
&= \sum_{t=0}^{\infty} \Pr_z[X_t = y, \tau_z^+ > t] - \Pr_z[X_t = y, \tau_z^+ > 0] + \sum_{t=1}^{\infty} \Pr_z[X_t = y, \tau_z^+ = t] \\
&= \left(\sum_{t=0}^{\infty} \Pr_z[X_t = y, \tau_z^+ > t] \right) - \Pr_z[X_0 = y, \tau_z^+ > 0] + \sum_{t=1}^{\infty} \Pr_z[X_t = y, \tau_z^+ = t] \\
&= \sum_{t=0}^{\infty} \Pr_z[X_t = y, \tau_z^+ > t] - \sum_{t=1}^{\infty} \Pr_z[X_t = y, \tau_z^+ = t] + \sum_{t=1}^{\infty} \Pr_z[X_t = y, \tau_z^+ = t] \\
&= \sum_{t=0}^{\infty} \Pr_z[X_t = y, \tau_z^+ > t] \\
&= \tilde{\pi}(y)
\end{aligned}$$

We normalize this with $\sum_{x \in \mathcal{X}} \tilde{\pi}(x)$ to get a probability distribution. Since $\sum_{x \in \mathcal{X}} \tilde{\pi}(x) = \mathbb{E}_z[\tau_z^+]$ (shown below), $\pi = \frac{\tilde{\pi}}{\mathbb{E}_z[\tau_z^+]}$ is a stationary distribution. (You should convince yourself that scaling $\tilde{\pi}$ by any scalar c still satisfies the stationarity condition i.e if $\tilde{\pi}P = \tilde{\pi}$ and $\pi = c \cdot \tilde{\pi}$ then $\tilde{\pi}P = \tilde{\pi} \Rightarrow \pi P = \pi$)

$$\begin{aligned}
\sum_{x \in \mathcal{X}} \tilde{\pi}(x) &= \sum_{x \in \mathcal{X}} \sum_{t=0}^{\infty} \Pr_z[X_t = x, \tau_z^+ > t] \\
&= \sum_{t=0}^{\infty} \sum_{x \in \mathcal{X}} \Pr_z[X_t = x, \tau_z^+ > t] && \text{(switching order of summations)} \\
&= \sum_{t=0}^{\infty} \sum_{x \in \mathcal{X}} \Pr_z[\tau_z^+ > t] \cdot \Pr_z[X_t = x] && \text{(independent events)} \\
&= \sum_{t=0}^{\infty} \Pr_z[\tau_z^+ > t] \\
&= \mathbb{E}_z[\tau_z^+]
\end{aligned}$$

□

We next argue that the stationary distribution is unique.

3 Proof of Uniqueness

Proposition 2. (Proof of uniqueness) The stationary distribution π of a finite, irreducible and aperiodic Markov Chain is unique.

Proof. Assume two distributions π_1 and π_2 that are both stationary distributions over $\mathcal{X}$. Consider a state $x \in \mathcal{X}$ that minimizes the ratio $\frac{\pi_1(x)}{\pi_2(x)}$. Let us call this minimizing value M . Then,

$$\pi_1(x) = \sum_{y \in \mathcal{X}} \pi_1(y) \cdot P(y, x) \geq \sum_{y \in \mathcal{X}} M \cdot \pi_2(y) \cdot P(y, x) = M \cdot \pi_2(x)$$

But we know that $\pi_1(x) = M \cdot \pi_2(x)$, so none of the $\pi_1(y) \geq M \cdot \pi_2(y)$ can be strict inequalities, i.e they must all be equalities. Therefore

$$\frac{\pi_1(x)}{\pi_2(x)} = M \text{ for all } x \in \mathcal{X}.$$

The following expansion shows that $M = 1$:

$$\begin{aligned}
\sum_{x \in \mathcal{X}} \pi_1(x) &= 1 && \text{(Sum of a probability distribution)} \\
\sum_{x \in \mathcal{X}} \frac{\pi_1(x)}{\pi_2(x)} \cdot \pi_2(x) &= 1 \\
\sum_{x \in \mathcal{X}} M \cdot \pi_2(x) &= 1 \\
M \sum_{x \in \mathcal{X}} \pi_2(x) &= 1 \\
M &= 1
\end{aligned}$$

Since $\pi_1(x) = \pi_2(x)$ for all $x \in \mathcal{X}$ (i.e $\pi_1 = \pi_2$), stationary distributions must be unique.

□

4 π as the reciprocal of expected first return time

Finally, our work above has left us with this handy identity:

Proposition 3. For the stationary distribution π of a finite irreducible and aperiodic Markov Chain,

$$\pi(z) = \frac{1}{\mathbb{E}_z[\tau_z^+]}$$

Proof. We saw that

$$\pi(z) = \frac{\tilde{\pi}(z)}{\mathbb{E}_z[\tau_z^+]}$$

Since $\tilde{\pi}(z) = 1$ (because the number of visits a random walk of the chain makes to z before returning to z is exactly 1 - and this gets logged at the exact moment the random walk returns to z), the proposition follows immediately. $\square$

5 Thinking through the assumptions

Let us next think about the assumptions we made, and why we they are reasonable. Recall that we have proved the existence and uniqueness of a stationary distribution for a *finite*, *irreducible* and *aperiodic* Markov Chain.

- *finite*: What if our chain was not finite? We couldn't formulate a transition matrix P , which is necessary for our proof. I believe a more general proof could remove the finiteness assumption and still prove existence and uniqueness of π . Our focus as such has been strictly in the discrete state space setting so we could formulate P .
- *irreducible*: What if our chain was not irreducible? Say A and B are disjoint subsets of $\mathcal{X}$ and $A \cup B = \mathcal{X}$. Then a sample path with $X_0 \in A$ would never get to B , and therefore could never "learn" the probability distribution of B . The stationary distribution is the long-term limiting distribution of the chain, and it should not depend *where* you start your chain from - but in the reducible case it is clear that the limiting distribution depends on where the chain starts. Therefore intuitively it makes sense that stationary distributions do not exist for reducible chains. If we were to use our proofs above, we could show the existence of stationary distributions (those distributions would only have probability mass over either A or B , depending on where X_0 started from). But we could not show uniqueness.
- *aperiodic*: Let us consider a periodic chain with two states A and B and $X_0 = A$. Then, the probability distribution at all time steps when t is odd is $\{\Pr[X_t = A] = 1, \Pr[X_t = B] = 0\}$ and when t is even is $\{\Pr[X_t = A] = 0, \Pr[X_t = B] = 1\}$. This means that the probability distribution never converges - we do not have a limiting distribution. Because of this issue, sometimes just called "periodicity issues" in the literature, we can prove the existence of π only for aperiodic chains.

References

- [1] David A. Levin and Yuval Peres. *Markov Chains and Mixing Times*. 2nd ed. American Mathematical Society, 2017.